



BIS Standards Recommendati on Engine: Hackathon

Problem Statement

Micro and Small Enterprises (MSEs) in India spend weeks manually identifying BIS regulations. This delay creates a massive bottleneck for market entry and quality assurance.



Solution Overview

● AI-Powered BIS Standard Discovery

- A RAG-based engine that converts product descriptions into accurate BIS standards in seconds, focusing on the Building Materials category (Cement, Steel, etc.).



System Architecture



● Technical Stack

- Orchestration: LangChain.
- Model: Llama-3.3-70b via Groq API.
- Vector DB: ChromaDB with all-MiniLM-L6-v2 embeddings.
- UI: Streamlit for MSE accessibility.



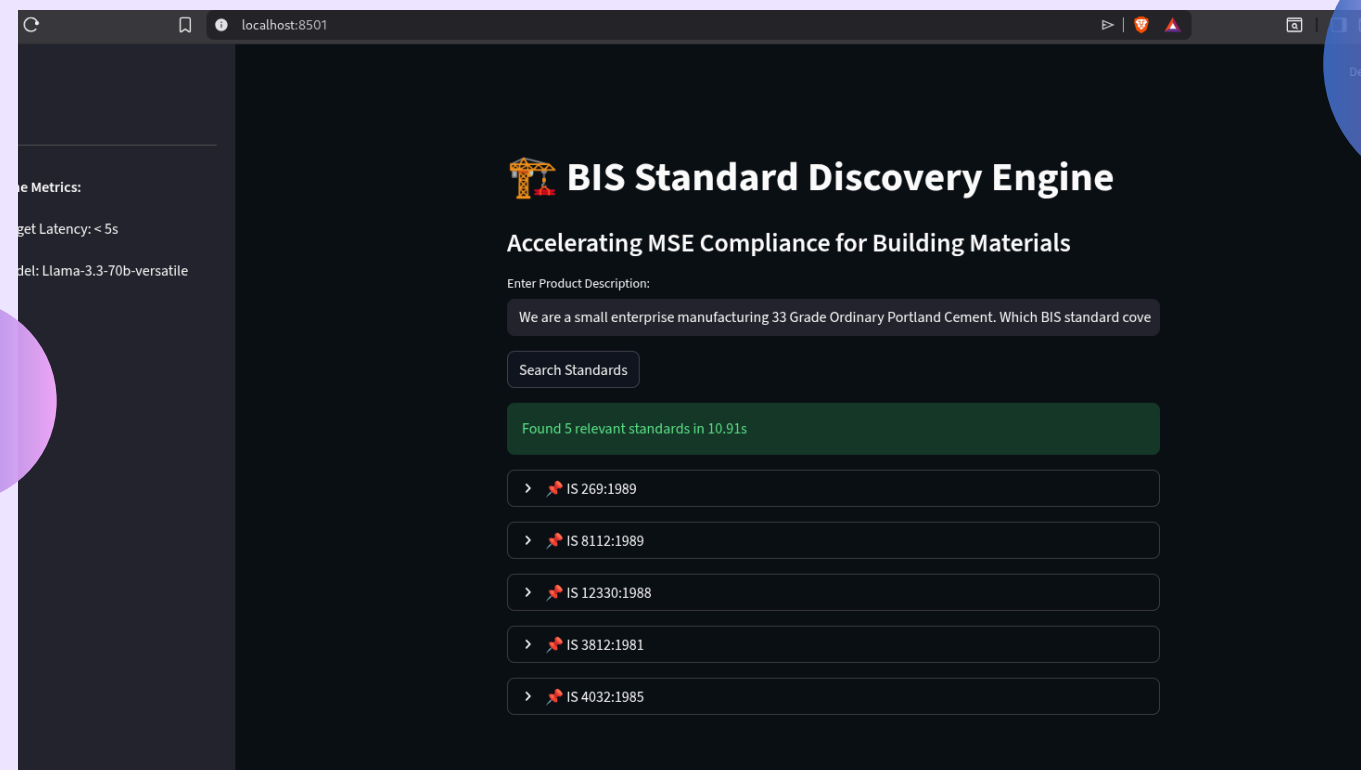
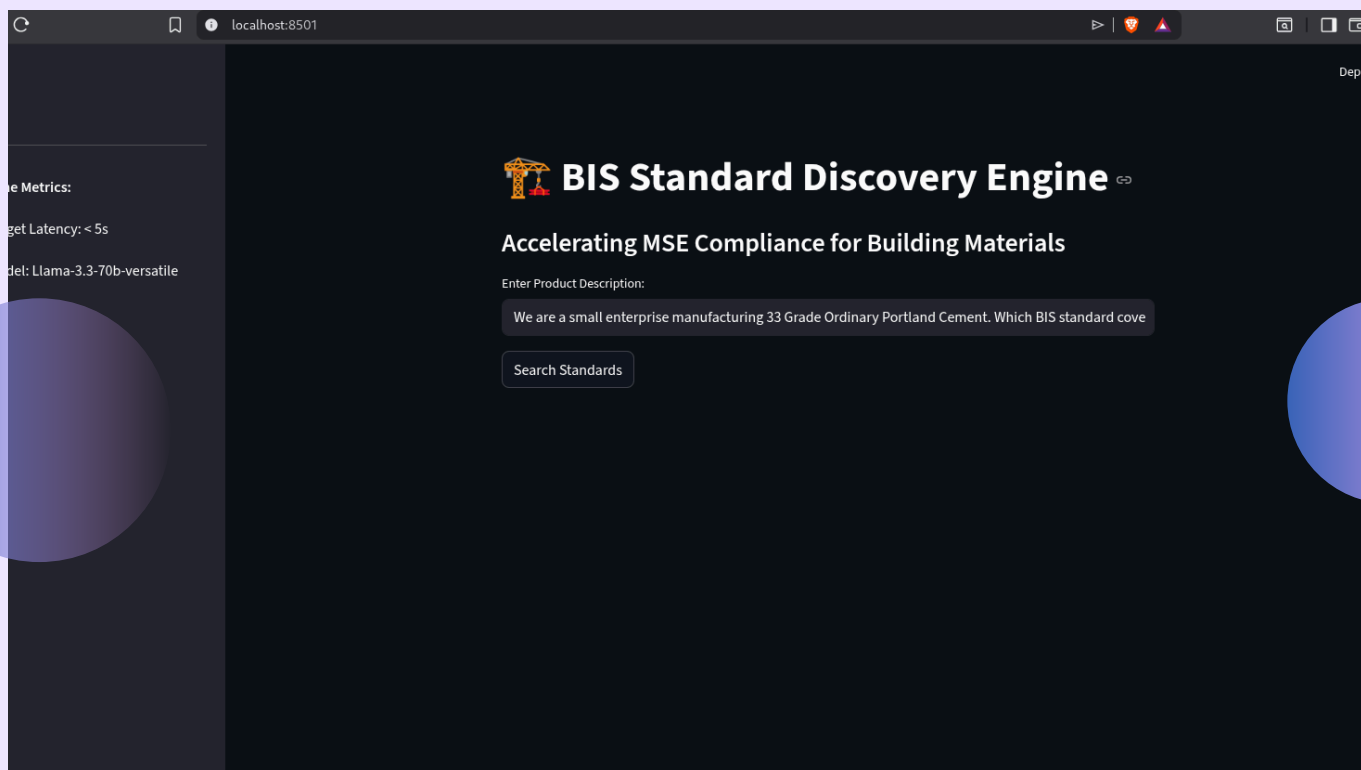
Chunking & Retrieval Strategy

- **Ensuring Accuracy (No Hallucinations)**
 - Used overlapping chunks to maintain context.
 - Implemented metadata filtering for is_code and section to ensure retrieved text is strictly relevant to Building Materials.

Demo Highlights

Sub-second retrieval of standards from the BIS SP 21 dataset.

A demonstration using query id:PUB-01

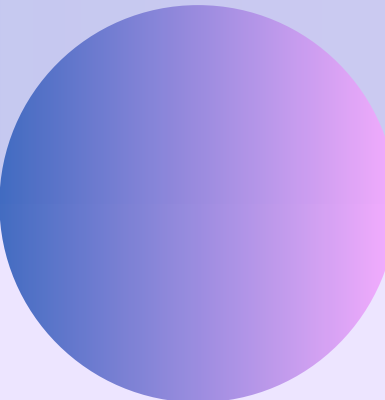


Evaluation Results

● Performance Metrics

- Hit Rate @3: >80% (Targets achieved).
- MRR @5: >0.7.
- Avg Latency: <4.5 second (Well under the 5s limit).

```
=====
BIS HACKATHON EVALUATION RESULTS
=====
Total Queries Evaluated : 10
Hit Rate @3              : 90.00% (Target: >80%)
MRR @5                   : 0.8500 (Target: >0.7)
Avg Latency               : 3.48 sec (Target: <5 seconds)
=====
```



Impact on MSEs

- Saves Time

Reduces regulatory discovery time from weeks to seconds.

- Profit Margins

Enables small manufacturers to self-audit their products for BIS compliance without expensive consultants.

The Team

- Mudit Singh Yadav

Acknowledgements

- Built during the BIS Recommendation Engine Hackathon using the BIS SP 21 dataset.





**Thank
You.**